

# Surrogate Modeling of Stokes Flow with Crystalline Inclusions

FNO vs. U-Net — Master's Thesis Defense

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# Outline

# Problem Statement — Crystal Sedimentation in Magma

## Physical Process

- Crystals (olivine, plagioclase) sediment in viscous magma
- Stokes regime:  $Re \ll 1$  (negligible inertia)
- Long-range interactions between crystals
- Influences thermal and chemical differentiation

## Governing Equations

$$\mathbf{0} = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0$$

## Computational Challenge

- LaMEM solver:  $\approx 8\text{--}9$  s per sample
- Scaling:  $\mathcal{O}(N^2)\text{--}\mathcal{O}(N^3)$  with crystal count  $N$
- Millions of degrees of freedom for  $N > 10$
- Systematic parameter studies barely feasible

## Goal

A fast surrogate model that outperforms the numerical solver by orders of magnitude

# Problem Statement — Stream Function as Learning Target

## Why the stream function $\psi$ ?

- Velocity field derived from  $\psi$ :

$$u_x = \frac{\partial \psi}{\partial z}, \quad u_z = -\frac{\partial \psi}{\partial x}$$

- Incompressibility ( $\nabla \cdot \mathbf{u} = 0$ ) satisfied by construction
- Smoother learning target than direct velocity components
- Physically consistent predictions by construction

## Input $\rightarrow$ Output

- **Input:** Geometry mask (crystal positions and radii on a  $256 \times 256$  grid)
- **Output:** Normalized stream function field  $\psi_{\text{norm}}$
- Velocity field derived via finite differences

## Error Metrics

$\bar{\epsilon}_\psi$ : mean relative  $L_2$  error on  $\psi$

$\bar{\epsilon}_v$ : mean relative  $L_2$  error on velocity

# Scope and Goals

## Research Question

Can neural surrogate models (FNO and U-Net) predict Stokes flow with multiple crystalline inclusions accurately and efficiently? Which architecture generalizes better?

### In Scope

- Two architectures: **FNO** and **U-Net**
- 4 experiments with increasing complexity
- Training & evaluation on LaMEM-generated data
- Hyperparameter search (lr, batch size)
- Out-of-distribution (OOD) generalization analysis

**FNO:** 100 epochs training

### Out of Scope

- 3D simulations
- Time-dependent flows
- Non-stationary boundary conditions
- Direct pressure field prediction
- Physics-informed loss terms

**U-Net:** 180 epochs training

# Architecture Comparison

## Fourier Neural Operator (FNO)

- Learns operator  $\mathcal{G} : a \mapsto u$  between function spaces
- Convolution in the spectral domain (FFT)
- Retains only  $k_{\max} = 16$  frequency modes
- Global, long-range dependencies
- Resolution-independent (by design)
- Particularly suited for elliptic PDEs (Stokes)

**Advantage:** Global coherence, fewer parameters

## U-Net

- Encoder-decoder with skip connections
- Local convolutional kernels (spatial locality)
- Multi-scale feature extraction
- Strong at geometric structures
- Composite loss: MSE + gradient term
- High computational efficiency (CPU-native)

**Advantage:** Spatial detail, faster inference

*Hypothesis: FNO better for long-range Stokes interactions; U-Net stronger at local geometry boundaries*

# Architectures — Key Differences

| Property             | FNO                               | U-Net                |
|----------------------|-----------------------------------|----------------------|
| Learning paradigm    | Operator (function space)         | Image-to-image       |
| Receptive field      | Global (spectral)                 | Local (convolution)  |
| Frequency processing | Selective ( $k \leq k_{\max}$ )   | Implicit (pooling)   |
| High-frequency       | Limited ( $k_{\max} = 16$ )       | Better (skip conn.)  |
| Boundary conditions  | Softly enforced                   | Directly in geometry |
| Inference time       | $\approx 687$ ms (incl. transfer) | $\approx 5.5$ ms     |
| Speedup vs. LaMEM    | $\approx 12\times$                | $> 1500\times$       |

## Spectral Limitation of FNO

The FNO discards Fourier modes with  $k > k_{\max} = 16$ . For flows with sharp boundary layers (large crystals) this leads to systematic approximation errors.

# Experimental Design — Overview

## Dataset (LaMEM-generated)

- Domain:  $[-1, 1]^2$ , grid  $256 \times 256$
- Fixed: crystal radius  $r = 0.05$
- 1 000 train / 100 validation / 10 evaluation (per stratum)
- Normalized stream function as target

## Hyperparameter Search (Exp. 1)

- LR:  $10^{-3}$ ,  $5 \times 10^{-3}$ ,  $10^{-4}$ ,  $5 \times 10^{-4}$
- Batch size: 8 and 16
- $\Rightarrow$  8 configurations per architecture

## Exp. 1 Single-Crystal Baseline

$n = 1$ , hyperparameter search

## Exp. 2 Multi-Crystal Generalization

Training:  $n \in \{1, \dots, 10\}$

OOD:  $n \in \{11, 12, 15, 20, 25\}$

## Exp. 3 Stress Test

Training:  $n \leq 25$

OOD:  $n \in \{35, 50, 60, 80, 84\}$

## Exp. 4 Crystal Size

OOD radii:  $r \in \{0.005, 0.5\}$

Variable-radius FNO:

$r \in \{0.02, 0.05, 0.08\}$



# Experiment 1 — Single-Crystal Baseline

**Setup:**  $n = 1$  crystal, FNO 100 epochs / U-Net 180 epochs

| Architecture (lr, bs)                      | $\bar{\epsilon}_{\psi}$ | $\bar{\epsilon}_v$ |
|--------------------------------------------|-------------------------|--------------------|
| FNO Exp-1.4 ( $5 \times 10^{-4}$ , bs 16)  | <b>1.62%</b>            | 3.44%              |
| FNO Exp-1.2 ( $5 \times 10^{-3}$ , bs 8)   | 2.07%                   | 4.13%              |
| U-Net Exp-1.2 ( $5 \times 10^{-3}$ , bs 8) | 5.50%                   | 8.50%              |
| U-Net Exp-1.3 ( $10^{-4}$ , bs 16)         | 13.5%                   | 23.1%              |

## Key Findings

- FNO achieves 1.62% — factor  $\approx 3.4\times$  better than best U-Net
- Batch size 16 better than 8 for FNO
- U-Net: bs 8 clearly better than bs 16
- No overfitting (training and validation curves closely aligned)
- No spatial dependence of FNO error on crystal position

*[Figure: Bar chart Exp-1 — all configurations]*

## Experiment 2 — Multi-Crystal Generalization

**Training:**  $n \leq 10$  crystals

**Best configurations from Exp. 1**

| Architecture | In-dist $\bar{\epsilon}_\psi$<br>(mean $n = 1-10$ ) | OOD $\bar{\epsilon}_\psi$<br>( $n = 25$ ) |
|--------------|-----------------------------------------------------|-------------------------------------------|
| FNO          | $\sim 5.5\%$                                        | $\sim 13.8\%$                             |
| U-Net        | $\sim 22.1\%$                                       | $\sim 26.8\%$                             |

**FNO error grows with  $n$**

(4% at  $n = 1 \rightarrow 9\%$  at  $n = 10$ )

### Key Findings

- FNO:  $\approx 4\times$  better in-distribution
- FNO:  $\approx 2\times$  better at OOD ( $n = 25$ )
- U-Net: best checkpoint often early (epoch 43)
- U-Net error barely sensitive to  $n$  (17–25% in-dist)
- FNO advantage at OOD smaller than in-dist

*[Figure: FNO vs. U-Net over  $n$ ]*

## Experiment 3 — Stress Test

**Training:**  $n \leq 25$  crystals

**OOD evaluation:**  $n = 35, 50, 60, 80, 84$

| Architecture | In-dist<br>$\bar{\epsilon}_\psi$ | OOD<br>$\bar{\epsilon}_\psi$ ( $n = 50$ ) |
|--------------|----------------------------------|-------------------------------------------|
| FNO          | $\sim 6.8\%$                     | 12.4%                                     |
| U-Net        | $\sim 23\%$                      | 21.5%                                     |

**Noteworthy:** U-Net overtakes FNO at  $n = 50$ !

### Key Findings

- FNO:  $\approx 3.4\times$  better in-distribution
- At  $n = 50$ : FNO 12.4% vs. U-Net 21.5%
- Extreme OOD ( $n > 60$ ): FNO 15–22%
- FNO advantage diminishes with extreme OOD
- U-Net: more stable OOD error than expected (local convolutions help with dense packings?)
- Best-checkpoint FNO: epoch 100 (no overfitting)

*[Figure: Generalization curves Exp. 2 and 3 comparison]*

## Experiment 4 — Crystal Size

**Fixed baseline:** Training on  $r = 0.05$ , OOD radii  $r \in \{0.005, 0.5\}$

| Model         | Eval radius           | $\bar{\epsilon}_\psi$ |
|---------------|-----------------------|-----------------------|
| FNO (fixed)   | small ( $r = 0.005$ ) | 12.1%                 |
| FNO (fixed)   | large ( $r = 0.5$ )   | <b>71.3%</b>          |
| U-Net (fixed) | small                 | $\approx 12\%$        |
| U-Net (fixed) | large                 | $\approx 35\%$        |
| FNO (var.)    | all radii             | 10.4%                 |

### Key Findings

- **FNO fails on large crystals!**  
Boundary-layer frequencies  $> k_{\max}$
- U-Net degrades more gracefully (skip conn. help)
- Small crystals: both similarly robust
- **Variable-radius FNO**  
( $r \in \{0.02, 0.05, 0.08\}$ ):  
10.4%  $\rightarrow$  catastrophic failure prevented
- Large std. dev. ( $\pm 10.8\%$ ): difficulty varies strongly

## Results — Overview

| Experiment             | Architecture | $\bar{\epsilon}_{\psi}^{\text{in-dist}}$ | $\bar{\epsilon}_{\psi}^{\text{OOD}}$ |
|------------------------|--------------|------------------------------------------|--------------------------------------|
| Exp. 1 ( $n = 1$ )     | FNO          | 1.62%                                    | —                                    |
|                        | U-Net        | 5.50%                                    | —                                    |
| Exp. 2 ( $n \leq 10$ ) | FNO          | $\sim 5.5\%$                             | 13.8% ( $n = 25$ )                   |
|                        | U-Net        | $\sim 22\%$                              | 26.8% ( $n = 25$ )                   |
| Exp. 3 ( $n \leq 25$ ) | FNO          | $\sim 6.8\%$                             | 12.4% ( $n = 50$ )                   |
|                        | U-Net        | $\sim 23\%$                              | 21.5% ( $n = 50$ )                   |
| Exp. 4 (size, fixed)   | FNO          | —                                        | 71.3% (large)                        |
|                        | U-Net        | —                                        | $\sim 35\%$ (large)                  |
| Exp. 4 (var. FNO)      | FNO          | —                                        | 10.4% (all radii)                    |
| Speedup vs. LaMEM      |              | $\approx 12\times$                       | $> 1500\times$                       |

# Results — FNO Dominance and Its Limits

## FNO Strengths

- Clearly better in-distribution (factor 3–4 $\times$ )
- Better OOD generalization to more crystals
- No overfitting over 100 epochs
- Monotone error convergence
- Spatially uniform error distribution

## U-Net Strengths

- 1500 $\times$  faster than LaMEM
- More robust on large crystals (Exp. 4)
- OOD error grows slower at extreme  $n$
- Easier to deploy on CPU

## Critical Limitations

- **FNO:** Spectral bias at  $k > k_{\max}$   
→ catastrophic failure on large crystals (71%)
- **U-Net:** Generally higher base error  
→ still behind despite 180 epochs
- Both:  $n > 50$  only limitedly reliable
- Very small evaluation sets ( $n = 10$  samples)

## Conclusion

FNO is the stronger architecture for this task, as long as no extreme geometry scales occur.

## Results — Qualitative Predictions

*[Figure: Predicted vs. ground-truth  $\psi$  and velocity field]*

*[Figure: Error field (residual) — FNO vs. U-Net]*

## Results — Generalization over Crystal Count

*[Figure: Exp. 2 and 3 —  $\bar{\varepsilon}_\psi$  over  $n$  for FNO and U-Net]*

*[Figure: Overview bar chart all experiments]*

**Exp. 2 → Exp. 3:** Both models improve with more training data ( $n \leq 10 \rightarrow n \leq 25$ )

**OOD boundary:** FNO advantage diminishes at  $n > 3 \times n_{\text{train,max}}$



# Results — Crystal Size and Spectral Bias

*[Figure: Exp. 4 — error by crystal size]*

## Interpretation

- Large crystal  $\rightarrow$  steep gradients near boundary
- FNO needs high Fourier modes ( $k > 16$ )  $\rightarrow$  fails
- U-Net: convolutional kernels capture local sharpness better
- Small crystal: smooth flow  $\rightarrow$  both robust

## Variable-Radius Solution

- Training on  $r \in \{0.02, 0.05, 0.08\}$
- $\bar{\epsilon}_\psi = 10.4\%$  — no catastrophic failure
- But: still higher than in-dist Exp. 1 (1.62%)
- Shows: size variability in training data is essential

## Key Contributions

- 1 Systematic FNO–U-Net comparison on Stokes flow with up to 25 crystals
- 2 FNO outperforms U-Net in-distribution by factor 3–4
- 3 Both models generalize to unseen crystal counts
- 4 Spectral bias of FNO identified and quantified
- 5 Variable-radius training as solution for size OOD
- 6 Computational savings:  $12\times$  (FNO) to  $> 1500\times$  (U-Net)

## Outlook

- Higher  $k_{\max}$  values or adaptive spectral modes
- 3D extension
- Physics-informed loss terms
- Larger evaluation sets (more robust statistics)
- Combining FNO + U-Net (U-FNO approach)
- Direct coupling with LaMEM workflow

# Thank you!

Questions and Discussion

**Contact:** paul.baselt@example.com  
**Code:** FNO\_Ansatz/ & UNET\_Ansatz/  
**Data:** Ergebnisse/

## Backup — FNO Architecture Details

- Fourier layer: Input  $\rightarrow$  FFT  $\rightarrow$  weights on  $k \leq k_{\max}$  modes  $\rightarrow$  IFFT  $+ W \cdot v$
- $k_{\max} = 16$ , 4 Fourier layers,  $d_v = 64$  channels
- Activation function: GELU
- Adam optimizer, 100 epochs, MSE loss
- Resolution:  $256 \times 256$  (resolution-independent by design)

## Backup — U-Net Architecture Details

- Encoder: 4 levels, each Conv-BN-ReLU, MaxPool
- Decoder: transposed conv + skip connections
- Composite loss:  $\text{MSE} + \lambda \cdot \|\nabla(\hat{\psi} - \psi)\|^2$
- Adam optimizer, 180 epochs
- Best-checkpoint selection via validation  $\bar{\epsilon}_{\psi}$

## Backup — Metrics in Detail

- $\bar{\varepsilon}_\psi = \frac{1}{N} \sum_{i=1}^N \frac{\|\hat{\psi}_i - \psi_i\|_2}{\|\psi_i\|_2}$ : mean relative  $L_2$  error on  $\psi$
- $\bar{\varepsilon}_v$ : analogous for derived velocity field
- $\bar{\varepsilon}_{\max}$ : mean maximum absolute error on  $\psi$
- $\overline{\text{MSE}}_\psi$ : mean squared error on  $\psi$
- Divergence RMS: physical sanity check ( $\sim 10^{-26}$  for FNO)

## Backup — Experiment 1 Full Table (FNO)

| Config (lr, bs)                       | $\bar{\epsilon}_\psi$               | $\bar{\epsilon}_v$ | $\overline{\text{MSE}}_\psi$ |
|---------------------------------------|-------------------------------------|--------------------|------------------------------|
| Exp-1.1 ( $10^{-3}$ , bs 16)          | $2.06 \pm 0.72\%$                   | $3.02 \pm 0.53\%$  | 0.0157                       |
| Exp-1.2 ( $5 \times 10^{-3}$ , bs 16) | $2.31 \pm 0.93\%$                   | $4.15 \pm 0.79\%$  | 0.0227                       |
| Exp-1.3 ( $10^{-4}$ , bs 16)          | $5.10 \pm 2.59\%$                   | $8.18 \pm 1.34\%$  | 0.1091                       |
| Exp-1.4 ( $5 \times 10^{-4}$ , bs 16) | <b><math>1.62 \pm 0.49\%</math></b> | $3.44 \pm 0.37\%$  | <b>0.0090</b>                |
| Exp-1.1 ( $10^{-3}$ , bs 8)           | $4.18 \pm 0.79\%$                   | $4.89 \pm 0.90\%$  | 0.0595                       |
| Exp-1.2 ( $5 \times 10^{-3}$ , bs 8)  | <b><math>2.07 \pm 0.65\%</math></b> | $4.13 \pm 0.56\%$  | 0.0164                       |
| Exp-1.3 ( $10^{-4}$ , bs 8)           | 5.85%                               | —                  | —                            |
| Exp-1.4 ( $5 \times 10^{-4}$ , bs 8)  | $2.61 \pm \dots \%$                 | 3.82%              | —                            |